Question:

Carlos is planting a lemon tree. The tree will cost $90 to plant. Each year it will grow 7 lemons, which he can sell for $1.5 each. It costs $3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<think>
the initial cost of planting the tree is $90
the tree earns 7 lemons per year, and each lemon is sold for $1 . 5, so the total earnings per year are 7 * 1 . 5 = 10 . 5 dollars
the cost of maintaining the tree per year is $3
to break even, the earnings should be greater than the cost, so we need to find the year when 10 . 5 - 3 > 0
this means that 7 . 5 > 0, which is always true, but we need to find the year when the tree starts earning money
let's calculate the profit per year, which is 10 . 5 - 3 = 7 . 5
since the profit is always positive, we need to find the year when the total profit is greater than the initial cost of planting the tree
let's calculate the total profit after n years, which is 7 . 5 * n
we need to find the smallest n such that 7 . 5 * n > 90
dividing both sides by 7 . 5, we get n > 12
since n is the number of years, it should be an integer, so the smallest n that satisfies the inequality is 13
</think><answer>13</answer>